Randomised Experiments and Covariates

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Preliminaries

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- PS1 submitted
- PS2 submitted
- Zoom recording links in `/admin/`

Random Assignment Mechanisms

- Simple/complete randomisation
  (Bernoulli trial, prob π)
- Complete randomisation / random allocation
  (fixed proportion to tr)
- Blocked randomisations
  (fixed proportion to tr, w/in group)
- Cluster randomisations
  (assignment at higher level)

Motivation: A Causal Inference Question

"Would a canvassing policy increase enrolment in a health insurance programme?"

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1	Dem		
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3	Rep		
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Suppose we observationally measure

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		Diff in Means: (Yes − No)	40

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Suppose we randomly assign 2 Tr, 2 Co, and measure

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Seriously? Well, ...

```r
x <- sample(0:1, size = 10, replace = TRUE)
```

```r
x
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1
```

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$$SE(\widehat{ATE}) = \sqrt{\frac{1}{N-1}\left[\frac{m\text{Var}(Y_i(0))}{N-m} + \frac{(N-m)\text{Var}(Y_i(1))}{m} + 2\text{Cov}(Y_i(0), Y_i(1))\right.}$$

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- If $\text{Var}(Y_i(0)) \neq \text{Var}(Y_i(1))$, allocate $\rightarrow$ higher-Variance condition

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$\widehat{ATE}_{+\text{cov}} = -7.5, 0, 7.5$

$\widehat{ATE}_{-\text{cov}} = 2.5, 0, -2.5$ (less variance!)

Summary

To minimise (parametric) variance/SE:

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- If equal variances, make $n_{Tr} = n_{Co}$
- If unequal variances, more units to higher variance condition (to a point)
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To minimise (parametric) variance/SE:

- If equal variances, make $n_{Tr} = n_{Co}$
- If unequal variances, more units to higher variance condition (to a point)
- If possible, define outcome s.t. $Cov(Y(1), Y(0)) < 0$

- Contrasting treatments: individs respond opposite directions
- Email A makes Dems donate a lot, Reps none
- Email B makes Reps donate a lot, Dems none

Example: Canvassing and Enrolment

But, is obs difference causal?

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What *would have* happened to "No" precincts if "Yes"?

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What do we really want to know?

Does canvassing actually *change* enrolment in precinct?
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What *would have* happened to "No" precincts if "Yes"?

What would have happened under *other* conditions?

Example: Canvassing and Enrolment

Suppose we can know both *potential outcomes* ...

Precinct	Party	Canvass?	Enrol % if No Canvass	Enrol % if Canvass
1	Dem	_	20	60
2	Dem	_	30	70
3	Rep	_	20	30
4	Rep	_	30	40
		Means:	25	50

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		Means:	25	50

$\text{ATE} = 50 - 25 = 25$

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4	Rep	_	30	40
		Means:	25	50

$\text{ATE} = 50 - 25 = 25$

(True or an estimate?)

Example: Canvassing and Enrolment

Another way to think about same information:

Precinct	Party	Canvass?	Enrol % if No Canvass	Enrol % if Canvass	True Precinct Effect
1	Dem	_	20	60	40
2	Dem	_	30	70	40
3	Rep	_	20	30	10
4	Rep	_	30	40	10
		Means:	25	50	25

$$\text{ATE} = (40 + 40 + 10 + 10)/4 = 25$$

The Fundamental Problem of Causal Inference

We can't observe both "Canvassed" and "Not Canvassed" for a precinct.

We can't observe both *potential outcomes* (*counterfactuals*).

So, how can we get a good causal estimate?

Example: Canvassing and Enrolment

Suppose we observe ...

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1	Dem	Yes		60
2	Dem	Yes		70
3	Rep	No	20	
4	Rep	No	30	
		Means:	25	65

Estimated ATE $= 65 - 25 = 40$

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Estimated ATE $= 65 - 25 = 40$ ☹ (too big)

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		Means:	30	45

Estimated ATE $= 45 - 30 = 15$

Example: Canvassing and Enrolment

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1	Dem	Yes		60
2	Dem	No	30	
3	Rep	Yes		30
4	Rep	No	30	
		Means:	30	45

Estimated ATE $= 45 - 30 = 15$ ☹ (too small; closer)

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In our random allocation, possible data were

Assignments	Est ATE
YYNN	40

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YYNN	40
NYNY	35
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NYYN	25
YNYN	15
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- $E(\hat{\bar{\tau}}) = \frac{1}{6}\cdot 40 + \frac{1}{6}\cdot 35 + \frac{1}{3}\cdot 25 + \frac{1}{6}\cdot 15 + \frac{1}{6}\cdot 10 = 25 = \bar{\tau}$

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Two assignments (YYNN and NNYY) leave treatment perfectly confounded with party.

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Let's ensure X does not predict T.

A Solution

Blocking:

Creating pre-treatmnt groups that look same on *predictors*.

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Blocking restricts possible data to

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Estimates have less variance, are closer to true ATE.

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Under random allocation, 5 possible estimates: 40, 15, 25, 35, 10.

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- Blocked pairs randomisation: $2^{\frac{n}{2}} = 2^{\frac{4}{2}} = 4$

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- (And, works better w/ bigger samples, fewer subgroups of interest)
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- **Can** observe X's, predictors of Y
- Check whether $T \stackrel{?}{\Rightarrow} X$

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- Covariate **balance**
- Estimate **closer to truth**
- Increased **efficiency**
- Triply-robust estimates: block, randomise, adjust
- Block-level effects
  $\rightsquigarrow$ different actors interested in different effects
- Guidelines for limited/uncertain resources

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Imai, King, and Stuart (2008):

Define *population average treatment effect*, PATE:

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and observed difference in sample means, D:

$$D \equiv \frac{1}{n/2} \sum_{i \in \{I_i = 1, T_i = 1\}} Y_i - \frac{1}{n/2} \sum_{i \in \{I_i = 1, T_i = 0\}} Y_i$$

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then, the total *estimation error* is

$$\Delta \equiv \text{PATE} - D$$

Why Block?

The *estimation error* can be decomposed:

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Blocking controls Δ_{T_X}, and, to the degree correlated, Δ_{T_U}.

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- More covariates: no exact comparable units (5 covars, 3 levels: $3^5 = 243$)
- More covariates often done informally

Multivariate, Continuous Blocking

Moore (2012)

- Like matching on PS, MD, start with dimension reduction

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Multivariate, Continuous Blocking

Moore (2012)

- Like matching on PS, MD, start with dimension reduction
- Like matching, select an algorithm
- Like matching, select other attributes (caliper, etc.)

Multivariate, Continuous Blocking

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- Create blocks of similar units
- Randomise within blocks

A Matrix of Distances

(First 5 rows of `x100`, variables `b1` and `b2`)

```
     1001 1002 1003 1004 1005
1001 0.00 2.68 2.51 2.46 1.36
1002 2.68 0.00 2.80 1.77 1.71
1003 2.51 2.80 0.00 1.09 1.53
1004 2.46 1.77 1.09 0.00 1.11
1005 1.36 1.71 1.53 1.11 0.00
```

Classes of Blocking Algorithms

- Optimal: consider all blockings; pick best.
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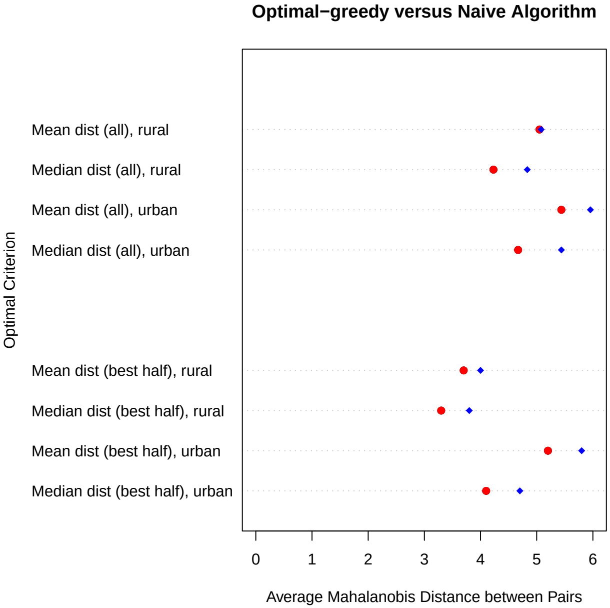
- Optimal: consider all blockings; pick best.
  (High-order problem)
- Optimal-greedy: consider all distances, pick best.
- Naïve greedy: Get best block for this unit now.

Which Multivariate Blocking Algorithm? (optimal greedy)



Deploying Limited Resources: opt-greedy algorithm

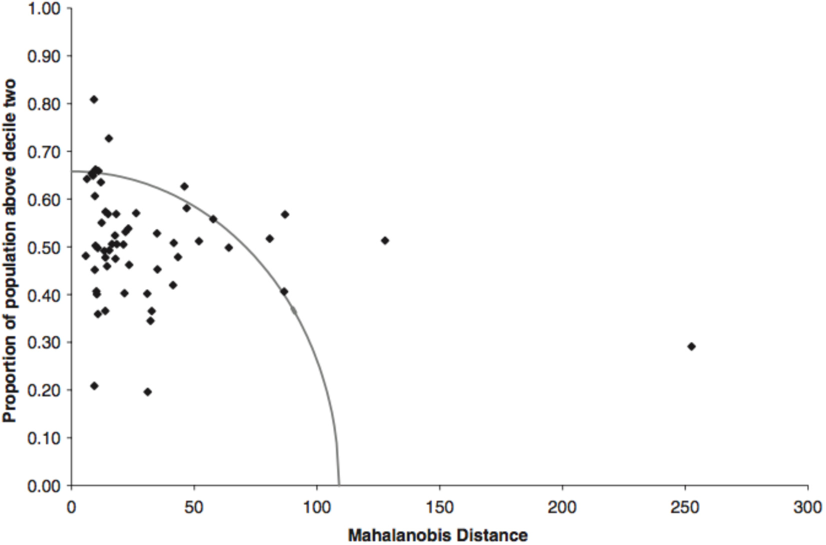


Figure 4. Choosing a subset of pairs for survey. We conducted our survey in 100 of

Why Block: Balance

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Simulation study: 100 units, $X_1 \sim N(0,1)$, $X_2 \sim \text{Unif}(0,1)$, $X_3 \sim \chi^2_2$; 1000 such experiments. Assg treatmnt in 3 ways. p-value from `xBalance`.

Why Block: Balance

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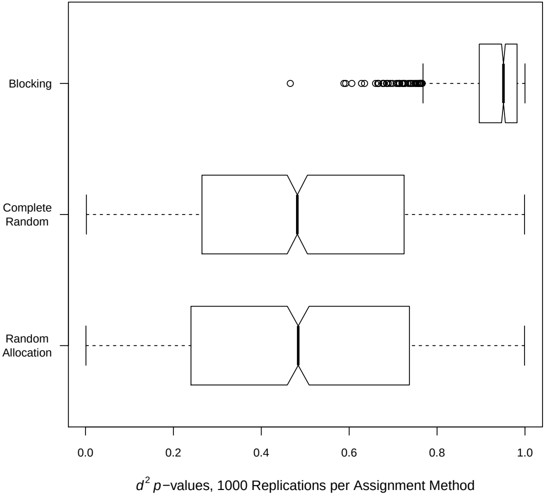

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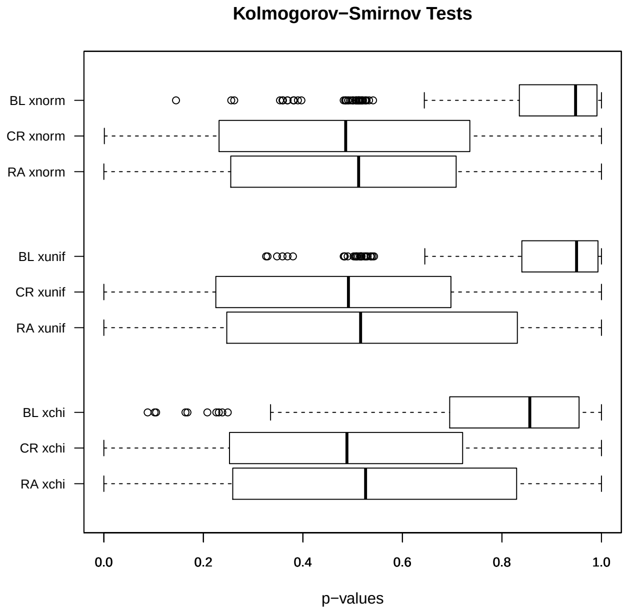
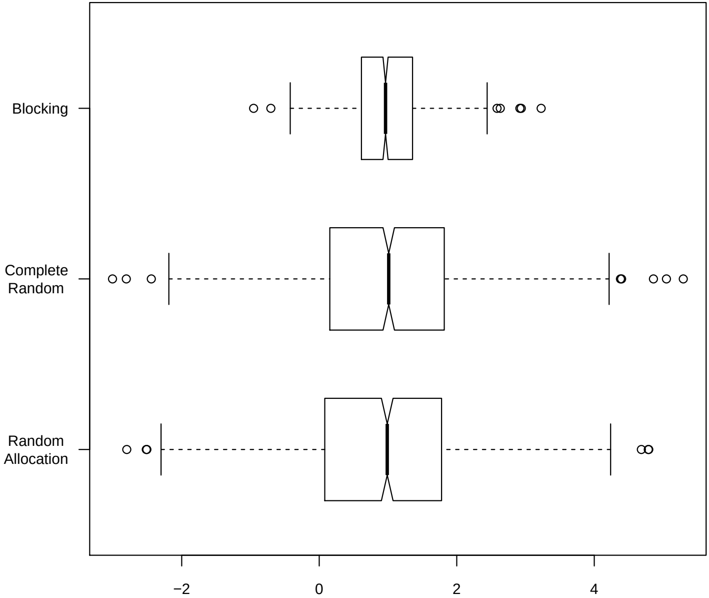
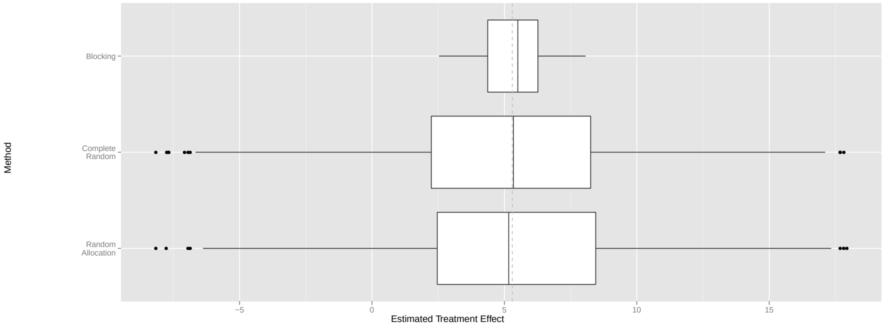


Why Block: Efficiency



Why Block: Efficiency under TE Heterogeneity



Applications: Seguro Popular, Perry Preschool, Ignatieff Campaign

Experiment: Randomised Health Infrastructure & Insurance

King et al. (2009)

- **Intervention**: resources for medical services, preventive care, pharmaceuticals, access, and financial health protection
- **Beneficiaries**: 50M Mexicans (half the pop) w/o access to health care

- **Cost** 2005 full +1% GDP new money annually
- One of **largest** health reforms of any country in 2 decades
- Most **visible** accomplishment of Fox administration
- Major **issue** 2006 pres. campaign (vote choice effect: 7-11% for PAN)

- **Randomised** 74/148 health clusters to first roll-out, infrastructure spending, encouragement to enroll
- But – do **better** than pure coin flip

Designing the *Seguro Popular* Experiment

How can we learn **most** from 148 cluster experiment?

- Ensure causes of HH expenditures evenly *balanced* betwn Tr and Co

  (otherwise, Tr might be poorer, or Co more insured)
- Make sure effect estimates as *precise* as possible

  (prefer estimate of "9 to 11%" to "−10 to 30%")

Designing the *Seguro Popular* Experiment

- Data on clusters:

cluster	population	educ years	doctors	nurses
MCSSA000364	3130	4.00	1	1
MCSSA000504	6492	5.11	1	1
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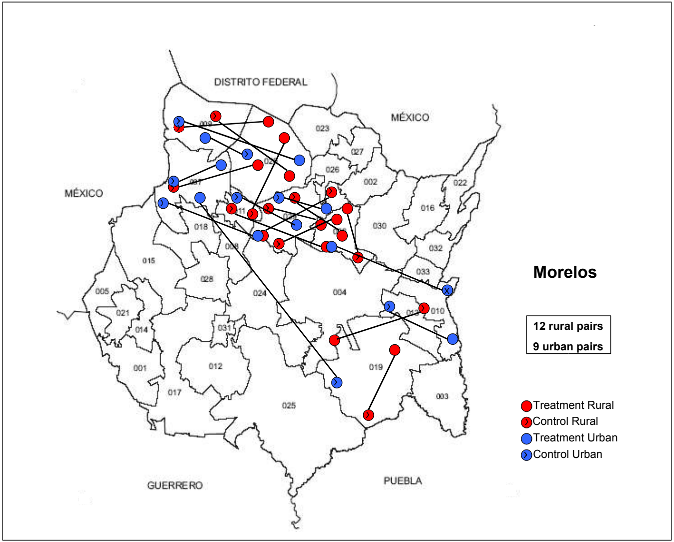
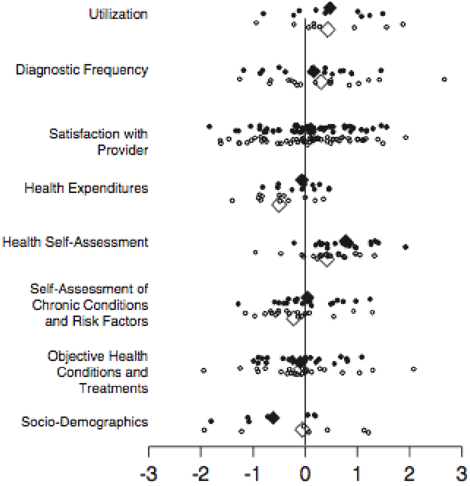
- Calculate how different clusters are from each other
- Mahalanobis distance between every pair of units:
  $MD_{ij} = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)'\hat{\Sigma}^{-1}(\mathbf{x}_i - \mathbf{x}_j)}$

Blocked Pairs, Morelos



Balance in Baseline Outcomes, *Seguro Popular*

King et al. (2007)



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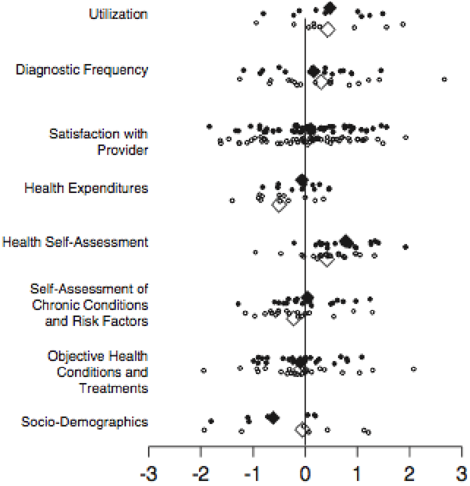
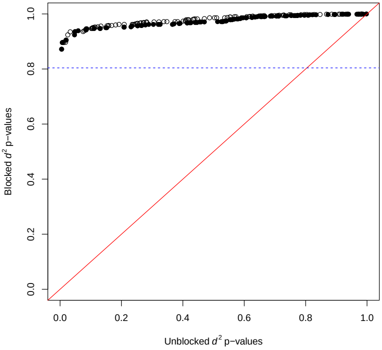


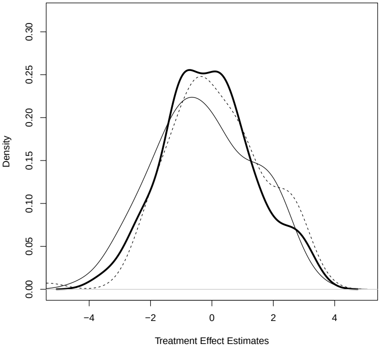
(Plus, SE's would have been $2\times$ to $6\times$ larger!)

Blocking in Applications: Balance and Efficiency

Moore (2012): Perry Preschool Experiment

Left: QQ plot of balance (100 blocked vs. unblocked)
Right: Est TE under sharp null (100 blocked vs. unblocked)

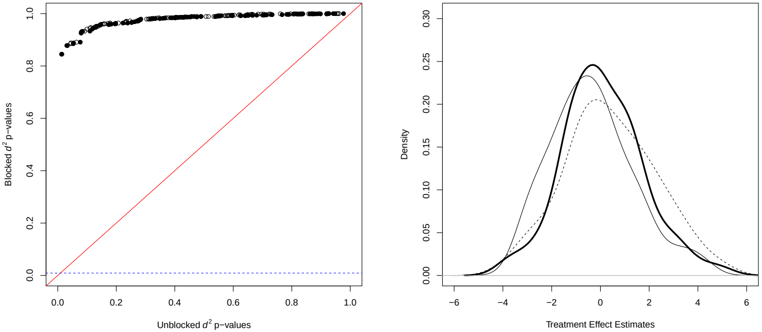




(SES, gender, IQ)

Balance in Applications: Balance and Efficiency

Considering more variables ...



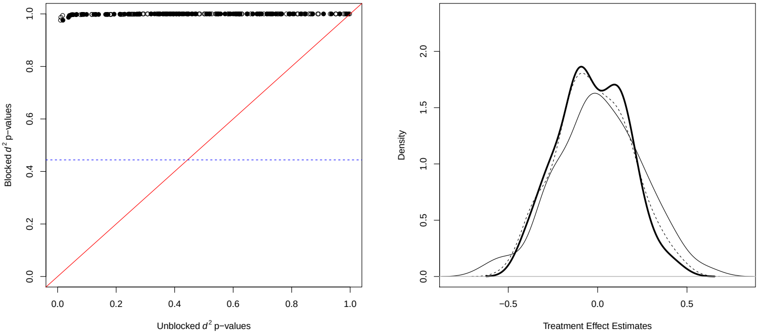
(+ siblings, AFDC, mom empl, educ, father, ...)

Balance in Applications

Loewen and Rubenson (2011): persuade party delegates to support Ignatieff

Left: QQ plot of balance (100 blocked vs. unblocked)

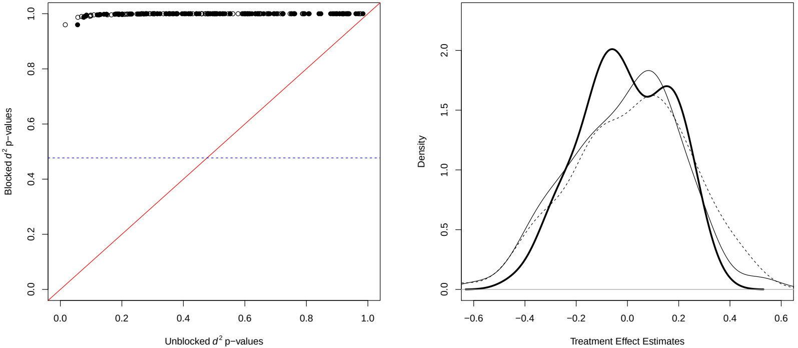
Right: Est TE under sharp null (100 blocked vs. unblocked)



(province, pledged, special constituency)

Balance in Applications

Considering more variables ...



(+ attention, interest)

Multivariate Continuous Blocking with blockTools

Moore and Schnakenberg (2025)

In R,

```r
library(blockTools)
data(x100)
head(x100)
```

```
    id id2  b1  b2 g  ig
1 1001 101 156 795 b 729
2 1002 102 813 469 a 627
3 1003 103 950 978 a 959
4 1004 104 991 781 a 661
5 1005 105 613 759 a 819
6 1006 106 654 838 b 643
```

Multivariate Continuous Blocking with blockTools

Block:

```
block.out <- block(data = x100, id.vars = "id",
                   block.vars = c("b1", "b2"))
```

Multivariate Continuous Blocking with `blockTools`

Block:

```
block.out <- block(data = x100, id.vars = "id",
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```

```
  Unit 1 Unit 2   Distance
1   1043   1040 0.01240000
2   1100   1020 0.02259275
3   1065   1027 0.02912651
4   1085   1081 0.03498815
5   1088   1061 0.04789253
```

Multivariate Continuous Blocking with blockTools

Assign:

```
assg.out <- assignment(block.out, seed = 157)
```

Multivariate Continuous Blocking with blockTools

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```

```
  Treatment 1 Treatment 2   Distance
1        1043        1040 0.01240000
2        1020        1100 0.02259275
3        1027        1065 0.02912651
4        1085        1081 0.03498815
5        1061        1088 0.04789253
```

Multivariate Continuous Blocking with blockTools

Diagnose:

```
diagnose(assg.out, data = x100, id.vars = "id",
          suspect.var = "b1", suspect.range = c(0,5))
```

Multivariate Continuous Blocking with `blockTools`

Diagnose:

```
diagnose(assg.out, data = x100, id.vars = "id",
         suspect.var = "b1", suspect.range = c(0,5))
```

Get block IDs

```
createBlockIDs(assg.out, data = x100, id.var = "id")
```

Multivariate Continuous Blocking with `blockTools`

Diagnose:

```
diagnose(assg.out, data = x100, id.vars = "id",
         suspect.var = "b1", suspect.range = c(0,5))
```

Get block IDs

```
createBlockIDs(assg.out, data = x100, id.var = "id")
```

Get balance:

```
assg2xBalance(assg.out, x100, id.var = "id",
              bal.vars = c("b1", "b2"))
```

Multivariate Continuous Blocking with `blockTools`

Extract conditions:

```
extract_conditions(assg.out, x100, id.var = "id")
```

```
 [1] 2 1 1 2 2 2 2 2 1 2 2 1 1 2 1 2 2 2 2 1 2 1 1 1 2 1
[38] 1 1 2 2 1 1 1 2 2 2 1 2 2 2 1 2 1 1 2 1 2 1 2 1 1 1
[75] 1 2 1 2 2 2 2 1 1 1 1 2 1 2 2 2 1 1 1 2 1 1 1 2 1 2
```

Multivariate Continuous Blocking with blockTools

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 [75] 1 2 1 2 2 2 2 1 1 1 1 2 1 2 2 2 1 1 1 2 1 1 1 2 1 2
```

```
x100 |> mutate(condition = extract_conditions(assg.out, x10
```

```
       id  id2  b1   b2 g  ig condition
1    1001 101 156  795 b 729         2
2    1002 102 813  469 a 627         1
3    1003 103 950  978 a 959         1
4    1004 104 991  781 a 661         2
5    1005 105 613  759 a 819         2
6    1006 106 654  838 b 643         2
7    1007 107 640  645 c  12         2
8    1008 108 681  404 a 221         2
```

Two Issues: Outliers and Interference

Covariate Weightings: Resistant Scaling

- $MD_{ij} = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)' \Sigma^{-1} (\mathbf{x}_i - \mathbf{x}_j)}$

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  - manual weighting

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- Resistant covariance estimators

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    1 constraint: include h points in interior

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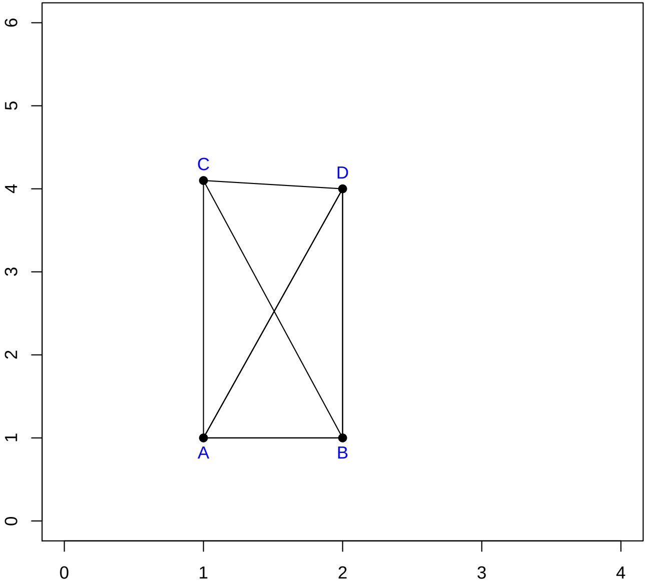
  - regular covariance
  - manual weighting
  - identity matrix (Euclidean dist)
- Resistant covariance estimators
  - MCD (Minimum Covariance Determinant)
    1 constraint: include h points in interior
  - MVE (Minimum Volume Ellipsoid)
    2 constraints: include h points, $p + 1$ points on boundary

Covariate Weightings: Resistant Scaling



Covariate Weightings: Resistant Scaling

Given outliers, use resistant estimates of Σ and blocks will stay *same*:

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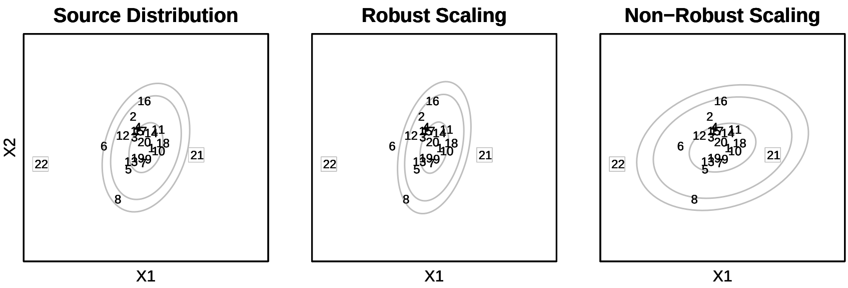
Given outliers, use resistant estimates of Σ and blocks will stay *same*:

		Distance from A		
Scale Matx	Points	B	C	D
Cov	A-D	**1.73**	1.76	2.45
Cov	A-$D + (1, 100)$	2.00	**0.08**	2.03
Cov	A-$D + (20, 100)$	1.72	1.01	**0.74**
MCD	A-D	**1.73**	1.76	2.45
MCD	A-$D + (1, 100)$	**1.73**	1.76	2.45
MCD	A-$D + (20, 100)$	**1.73**	1.76	2.45
MVE	A-D	**1.73**	1.76	2.45
MVE	A-$D + (1, 100)$	**1.73**	1.76	2.45
MVE	A-$D + (20, 100)$	**1.73**	1.76	2.45

(**Bold** = closest to A)

Covariate Weightings: Resistant Scaling



Covariate Weightings

Baseline: Exclude outliers, regular MD

	Unit 1	Unit 2	Distance
1	15	4	0.18
2	9	7	0.37
3	13	5	0.37
4	18	10	0.43
5	17	3	0.44
6	14	11	0.50
7	20	1	0.65
8	16	2	0.95
9	12	6	1.32
10	19	8	2.12

Covariate Weightings

Include outliers, resistant MD: Same blocks! ☺

	Unit 1	Unit 2	Distance
1	15	4	0.18
2	13	5	0.37
3	9	7	0.42
4	18	10	0.46
5	17	3	0.50
6	14	11	0.59
7	20	1	0.76
8	16	2	1.03
9	12	6	1.56
10	19	8	2.18
11	22	21	13.43

Covariate Weightings

Include outliers, non-resistant MD: Blocks shuffled by outliers ☹

	Unit 1	Unit 2	Distance
1	17	15	0.12
2	9	7	0.24
3	19	13	0.25
4	14	11	0.28
5	10	1	0.33
6	12	3	0.45
7	4	2	0.58
8	20	18	0.69
9	8	5	1.45
10	22	6	2.24
11	21	16	3.64

Interference

- We worry that effects may spillover: *interference*
- But nearby units are similar!
- Restrict blocks to units in a range

Multivariate Continuous Blocking with `blockTools`

Other arguments to `block()`

- `vcov.data`
- `n.tr`
- `algorithm`
- `distance`: `mahalanobis`, `mcd`, `mve`
- `weight`
- `level.two`: block states by most similar cities
- `valid.var`, `valid.range`: Goldilocks
- `seed`: (for `mcd` and `mve`)

A Note on Attrition

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  - Variance estimator unbiased
  - Easier to interpet as causal
- See, e.g., Fukumoto (2022)

Blocking for Sequential Experiments

What about Sequential Experiments?

Moore and Moore (2013)

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Moore and Moore (2013)

tall, male,
senior, biology

What about Sequential Experiments?

Moore and Moore (2013)

Have info!

Common Discrete Covariate-Adaptive Allocations

Biased coins

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Biased coins

Define unique covariate profile $\mathbf{p}$

Common Discrete Covariate-Adaptive Allocations

Biased coins

Define unique covariate profile $\mathbf{p}$

use

$$Pr(t = T) = \begin{cases} \frac{2}{3} & \text{if } n_{\mathbf{p}T} < n_{\mathbf{p}C} \\ \frac{1}{3} & \text{if } n_{\mathbf{p}T} > n_{\mathbf{p}C} \end{cases}$$

Common Discrete Covariate-Adaptive Allocations

Minimisation

Common Discrete Covariate-Adaptive Allocations

Minimisation

Given $\mathbf{p}$, score each variable s_j, $\left(\frac{n_{\mathbf{p}C}-n_{\mathbf{p}T}}{n_{\mathbf{p}C}+n_{\mathbf{p}T}+1}\right)$
combine (sum), rank conditions, assign π's

Common Discrete Covariate-Adaptive Allocations

Minimisation

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combine (sum), rank conditions, assign π's

	Gender		Age	
	M	F	Young	Old
Control	1	3	3	3
Treatment	2	4	3	1
s_j (Old M)	$-\frac{1}{4}$			$\frac{2}{5}$

Common Discrete Covariate-Adaptive Allocations

Minimisation

Given $\mathbf{p}$, score each variable s_j, $\left(\frac{n_{\mathbf{p}C} - n_{\mathbf{p}T}}{n_{\mathbf{p}C} + n_{\mathbf{p}T} + 1}\right)$
combine (sum), rank conditions, assign π's

	Gender		Age	
	M	F	Young	Old
Control	1	3	3	3
Treatment	2	4	3	1
s_j (Old M)	$-\frac{1}{4}$			$\frac{2}{5}$

(bias toward Treatment)

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Perfect balance?

	Gender		Age	
	M	F	Young	Old
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Perfect imbalance?

	Young M	Old M	Young F	Old F
Control	3	0	0	3
Treatment	0	3	3	0

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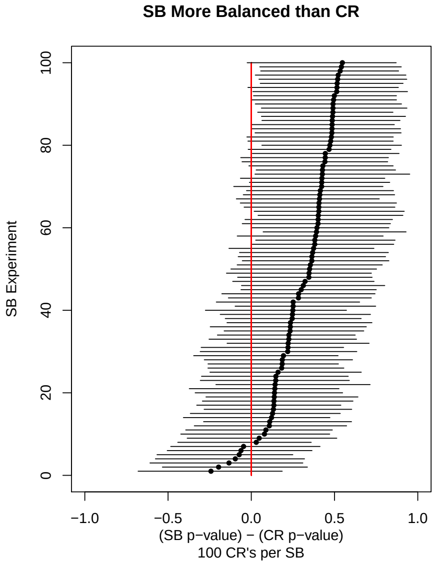
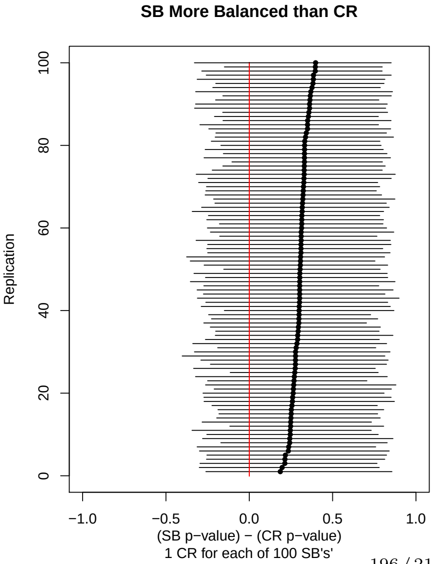
- Implement in actual sequential trial

Seq. Blocking More Balanced than CR

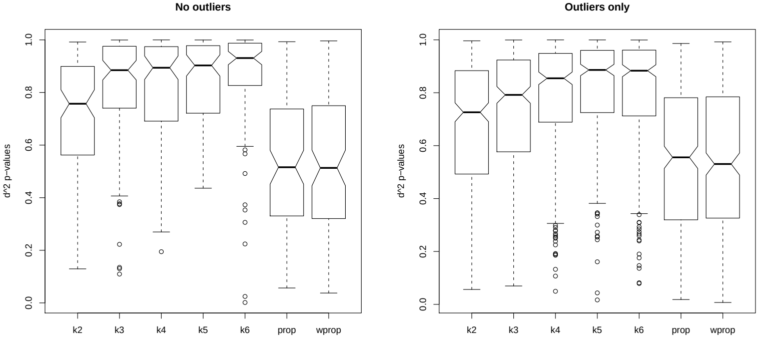
Correlated MVN





Selecting a Method for Experiments

Simulate (48 units, realistic means, SDs, bivariate corrs for 10 covariates) $\times 100$



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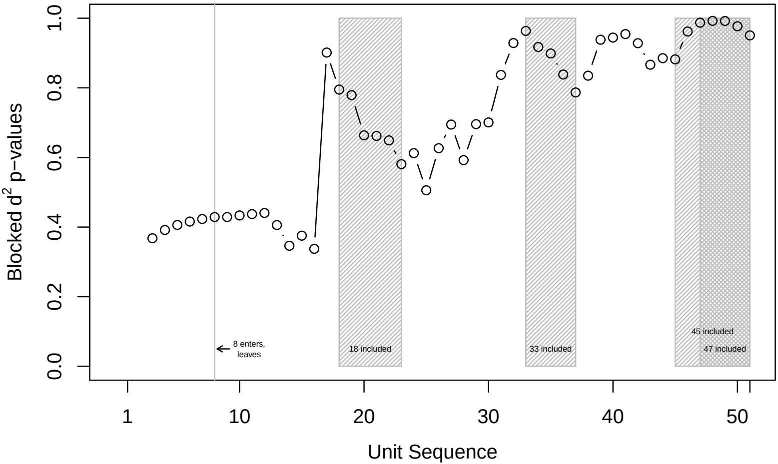
- 52 assigned total; 46 assigned Nov 09 - Jan 11; 39 follow-up.
- Two Conditions (daily practice)
  - `memory`: retrieve specific memories of life events in response to cue words
  - `anagrams`: rearrange letters of cue words to create new words (control)

Inputting Data in Real Time: Generalized Query

Unit 1

```
> seqblock1(query=T)
How many identification variables are there?
> 1
Enter the name of ID variable 1 without quotation marks.
> id
Enter the value of 'id'.
> 10624
How many exact blocking variables are there?
> 0
How many blocking variables are there?
> 2
Enter the name of blocking variable 1.
> x1
Enter the value of `x1'.
> 100
Should `x1' be restricted to certain values? [n/y]
> no
Enter the name of blocking variable 2.
> x2
Enter the value of `x2'.
> 80
How many experimental/treatment conditions are there?
> 2
```

Balance: Moore & Moore 2013



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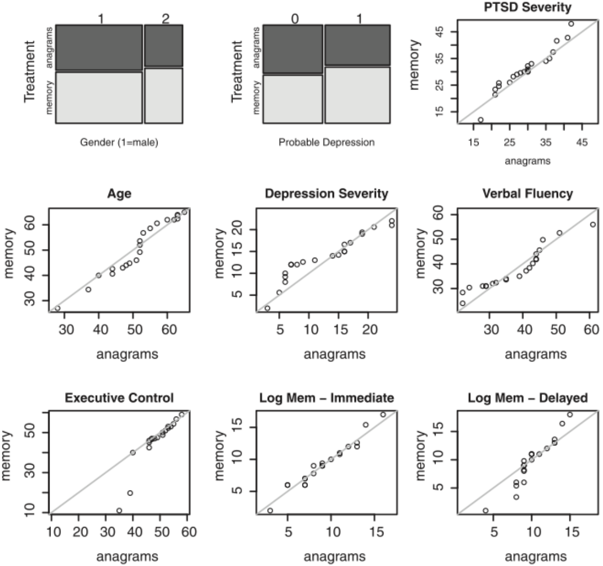

RI Confidence Intervals from blockTools

Get RI confidence interval from sequentially blocked data:

```
set.seed(407357912)
df <- tibble(id = 1:50,
             prov = sample(1:2, 50, replace = TRUE),
             age = sample(seq(18, 55, 1), 50, replace =
             treat = sample(0:1, 50, replace = TRUE),
             yobs = treat + sample(15:20, 50, replace =
```

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```

Check summary statistics:

```
df |> group_by(treat) |> summarise(m_yobs = mean(yobs))
```

```
# A tibble: 2 x 2
  treat m_yobs
  <int>  <dbl>
1     0   17.7
2     1   18.3
```

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Get RI confidence interval from sequentially blocked data:

```
invertRIconfInt(df,
                outcome.var = "yobs", tr.var = "treat",
                tau.abs.min = -2, tau.abs.max = 3,
                tau.length = 20, n.sb.p = 200,
                id.vars = "id", id.vals = "id",
                exact.vars = c("prov", "age"), exact.vals =
```

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```

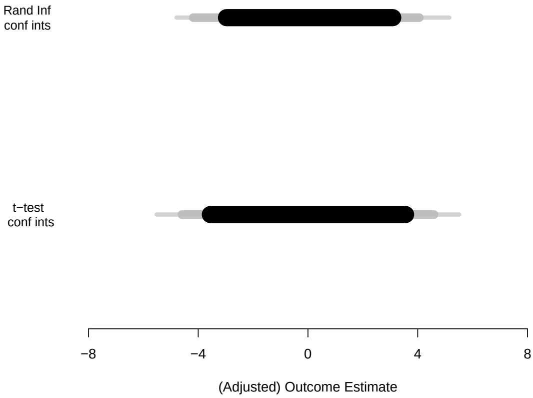
```
$ci95
[1] -0.16  0.11  0.37  0.63  0.89  1.16  1.42

$ci90
[1] -0.16  0.11  0.37  0.63  0.89  1.16

$ci80
[1] 0.11  0.37  0.63  0.89  1.16
```

RI Confidence Intervals: Moore & Moore 2013

Blocked estimates 9-15% more efficient



Analysis for Blocked Designs

If assignment probability for i varies by block j, let $p_{ij} = m_{ij}/N_j$. Weight each obs with

$$w_{ij} = \frac{d_i}{p_{ij}} + \frac{1 - d_i}{1 - p_{ij}}$$

Testing for Blocked Designs

‘As ye randomise so shall ye analyse’

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If blocks used to randomise, incorporate into RI.

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If blocks used to randomise, incorporate into RI.

1. Use actual blocks created, when possible
2. Reassign hypothetical treatment 1000 times
3. Obtain distribution of ATEs under (e.g.) sharp null
4. Compare observed ATE to randomisation distribution

Next:
Clustered Designs, Regression and Experiments
PS3 due today

References I

Fukumoto, Kentaro. 2022. "Nonignorable Attrition in Pairwise Randomized Experiments." *Political Analysis* 30 (1): 132–41.

Imai, Kosuke, Gary King, and Elizabeth A. Stuart. 2008. "Misunderstandings Between Experimentalists and Observationalists about Causal Inference." *Journal of the Royal Statistical Society, Series A* 171 (2): 481–502.

King, Gary, Emmanuela Gakidou, Kosuke Imai, Jason Lakin, Ryan T. Moore, Clayton Nall, Nirmala Ravishankar, et al. 2009. "Public Policy for the Poor? A Randomised Assessment of the Mexican Universal Health Insurance Programme." *The Lancet* 373 (9673): 1447–54.

King, Gary, Emmanuela Gakidou, Nirmala Ravishankar, Ryan T. Moore, Jason Lakin, Manett Vargas, Martha María Téllez-Rojo, Juan Eugenio Hernández Ávila, Mauricio Hernández Ávila, and Héctor Hernández Llamas. 2007. "A 'Politically Robust' Experimental Design for Public Policy Evaluation, with Application to the Mexican Universal Health Insurance Program." *Journal of Policy Analysis and Management* 26 (3): 479–509.

References II

Loewen, Peter John, and Daniel Rubenson. 2011. "For Want of a Nail: Negative Persuasion in a Party Leadership Race." *Party Politics* 17 (1): 45–65.

Moore, Ryan T. 2012. "Multivariate Continuous Blocking to Improve Political Science Experiments." *Political Analysis* 20 (4): 460–79. https://doi.org/10.1093/pan/mps025.

Moore, Ryan T., and Sally A. Moore. 2013. "Blocking for Sequential Political Experiments." *Political Analysis* 21 (4): 507–23.

Moore, Ryan T., and Keith Schnakenberg. 2025. *blockTools: Blocking, Assignment, and Diagnosing Interference in Randomized Experiments.* https://doi.org/10.32614/CRAN.package.blockTools.